Theory of Computation

Submission link: https://forms.gle/9QfcWiqewGmCcvG36

Exercise 8: (Context-free grammar part 1)

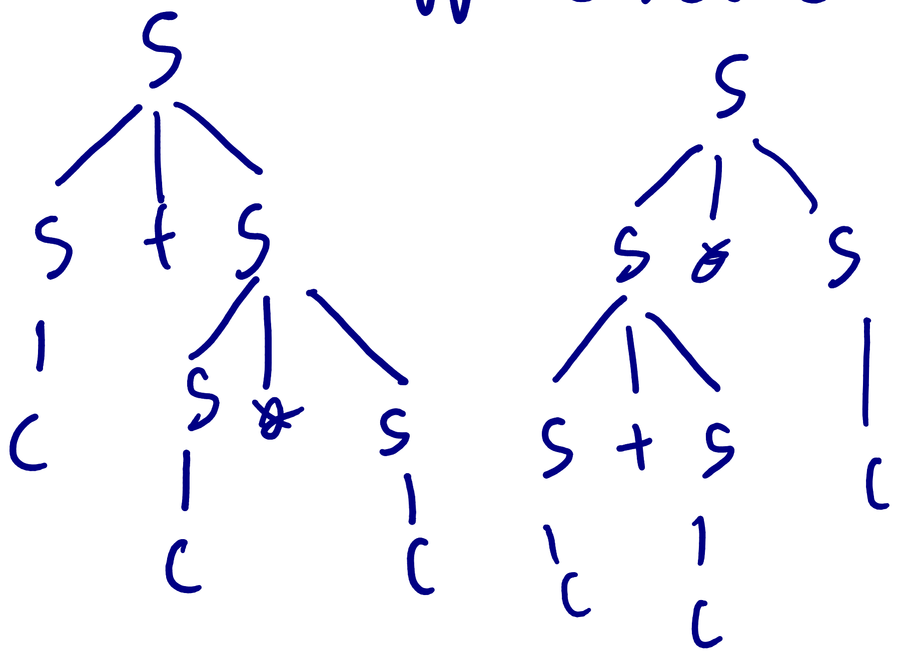
1. Prove that the following grammar is ambiguous.

$$S \to S + S \mid S - S \mid S * S \mid S/S \mid c$$

$W = c+c*c$

1) $W \in L(G)$

2) Find atleast 2 different derivation trees



2. Find CFG for the language L.

L $=\{ a^i b^j : i \leq j \}$

$S \to AB$

$A \to aAb \mid \lambda$

$B \to bB \mid \lambda$

$S \to aSb \mid bS \mid \lambda$

*3. Find the language of the following grammar.

$$G: S \rightarrow aA \mid bA \mid a \mid b$$
$$A \rightarrow aS \mid bS$$

$L(G) = \{a, b, aaa, aab, aba, abb, baa, bab, bba, bbb, aaaaa \ldots\}$

$L(G) = \{w \in \{a,b\}^* : |w| \text{ is odd}\}$

4) Find CFG for L.

$L = \{ a^i b^j c^k : k = i+j \}$

$S \rightarrow aSc \mid B \mid \lambda$

$B \rightarrow bBc \mid \lambda$